Haptic Feedback in Robotic-Surgery Training

Hypotheses

- **H1 (training-phase arousal).** Haptic feedback reduces the *rate* at which physiological/oculomotor arousal builds up across repetitions and with increasing task difficulty, relative to no haptic feedback — tested as Haptic × Repetition and Haptic × Level interactions.
- **H2 (transfer).** The benefit of training with haptic feedback persists when it is removed on a transfer task (elastic-band task, Baseline1 → Baseline2) — tested as a Phase × Group interaction.
- **H3 (carry-over control).** A structurally different task (Peg Transfer, Test1 → Test2) shows a smaller or absent version of the same interaction, distinguishing a general practice effect from a haptic-specific one.
- **H4 (performance).** Completion time and collision count show the same Haptic × Repetition / Haptic × Level pattern predicted for physiology in H1.
- **H5 (mechanism).** Physiological/oculomotor arousal statistically mediates part of haptic feedback's effect on performance.

2. Study Design and Data

2.1 Participants and phases

N = 47 novice participants were randomized to haptic feedback during training (H, n = 23) or no haptic feedback (N, n = 24). Haptic feedback, where present, was active only during the training phase; both transfer probes were performed without haptic feedback in both groups, isolating a retained effect of having trained with feedback from an online effect of currently receiving it.

Phase	Structure	Trials/subject	Purpose
Training	5 difficulty Levels × 10 repetitions each	50	Primary learning-rate test (H1/H4)
Baseline1 → Baseline2	Elastic-band task, pre/post training, 5 levels each	5 + 5	Transfer test (H2)
Test1 → Test2	Peg Transfer, pre/post training, single continuous task	1 + 1	Carry-over control (H3)

Repetition (1–10) and Level (1–5) are two separate, fully crossed design factors during training, not one factor — every training-phase model includes both explicitly.

2.2 Participant demographics

Demographics were collected via a post-study survey (LimeSurvey export, 49 respondents; two respondents, s00H and s01H, are pilot subjects never part of the analyzed 47 and are excluded from the table below).

Group	N	Age: mean ± SD (range)	Sex	Gaming frequency (Never/Rarely/Occ./Freq.)	Robot experience (None/Occ./Prof.)
N (non-haptic)	24	25.6 ± 2.6 (21–30)	14 M / 10 F	6 / 8 / 8 / 2	12 / 7 / 5
H (haptic)	23	28.0 ± 5.8 (21–47)	14 M / 9 F	3 / 7 / 4 / 9	12 / 7 / 4

Groups are balanced on sex and prior robotic experience. H skews slightly older (driven largely by one 47-year-old participant) and reports playing video games more frequently than N. Both age and gaming experience were included in the demographic-covariate sensitivity check (Section 4.8): the largest coefficient shift across all primary models was 3.6%, well under the pre-registered 20% material-change threshold, and no result's significance status changed — the unadjusted models stand as primary.

2.3 Data quality and exclusions

Every exclusion or missing-data mechanism below was verified against the actual pipeline source code and, where noted, against a live census of the current data (data_quality_audit.m).

Subject-level coverage

All 47 canonical subjects have at least one row in every one of the four feature tables — there are no full-subject dropouts. Row-level (per trial) coverage varies by modality and phase:

Table	Baseline rows / possible	Training rows / possible	Test rows / possible
EDA (GSR)	425 / 470 (90%)	2213 / 2350 (94%)	**77 / 94 (82%)**
Eye-tracking	452 / 470 (96%)	2305 / 2350 (98%)	92 / 94 (98%)
Performance	463 / 470 (99%)	2349 / 2350 (>99%)	94 / 94 (100%)
Force/vibration	463 / 470 (99%)	2349 / 2350 (>99%)	94 / 94 (100%)

"Possible" = 47 subjects × the number of acquisitions in that phase (5 baseline levels, 50 training reps, 2 test trials per subject-pair, counted per phase-half). GSR coverage is the outlier, and specifically lowest at Test (82%): the GSR row-drop rule below removes the entire trial on a single dropout gap, and the Peg Transfer test trial is one long continuous recording rather than five short per-level segments, so it has more opportunity to contain one disqualifying gap. This lower coverage appears in both groups roughly evenly (17 N / 16 H usable pairs at Test in the Layer1 comparisons, Section 4.1.2) and is not a haptic-condition-related artifact.

Row-level exclusion rules (active, verified in source)

Mechanism	Rule	Effect
GSR dropout gap	Trial dropped if any gap > cfg.max_gap_s = 0.50 s, or fewer than 20 samples in the trial window, or cvxEDA decomposition fails	Row removed entirely from eda_features.mat — never interpolated across
Eye-tracking dropout gap	Trial dropped if any gap > cfg.eye_max_gap_s = 0.20 s, or fewer than 20 samples	Row removed entirely from eye_features.mat
Eye-tracking blinks	Individual samples where the tracker's own blink flag = 1	Excluded within an otherwise-surviving trial, not interpolated
Force/vibration sensor missing	force.mat or accel.mat absent for that acquisition	tssf_raw / tssv set to NaN for that row; row itself is kept
Collision detection: corrupted audio	Remaining audio span too short to run peak detection after removing non-monotonic timestamps	Acquisition skipped entirely (no collision_results.mat saved) — 8 acquisitions total, listed below

NaN rates and what drives them

Outcome	NaN rate	Cause
scl_mean / scr_freq / scr_mean_amp / duration_s / n_collisions	0.0%	No missing rows in the surviving tables for these outcomes
collision_scr_amp_mean	9.6% (261/2715)	Structural — NaN when a trial had zero collisions (nothing to lock the response to), spread across 44/47 subjects; not a data defect
collision_pupil_response_mean	10.8% (307/2849)	Same structural cause as above, spread across 45/47 subjects
pupil_mean / pupil_slope_norm / gaze_dispersion / saccade_rate	0.2% (6/2849)	Concentrated in 3 subjects (s12H, s30N, s31H) — a handful of individual trial dropouts, negligible

Outcome	NaN rate	Cause
tssf_raw	4.4% (128/2906)	97% of these rows (124/128) come from exactly two subjects, s03H and s04H, who are missing force-sensor data for essentially their entire session (62/63 acquisitions each) — a hardware/connection dropout specific to those two sessions, not a task- or condition-related pattern
tssv	0.1% (4/2906)	The same 4 corrupted-audio acquisitions listed below — a shared-infrastructure coincidence

Recommendation carried into every model using tssf_raw: report n and flag that s03H/s04H contribute no force-sensor observations, rather than silently reducing the effective H-group n from 23 to 21 without comment.

Collision detection coverage

Collision detection succeeded (produced a collision_results.mat) for 2900 of 2908 eligible acquisitions with audio recorded (99.7%). The 8 skipped acquisitions, all attributable to corrupted/near-empty audio after non-monotonic-timestamp removal:

subject_s44N/Baseline1/Level3, subject_s44N/Baseline2/Level1, subject_s23H/level_L5/rep_01, subject_s29H/level_L5/rep_05, subject_s33H/level_L2/rep_07, subject_s35H/Baseline1/Level1, subject_s35H/level_L5/rep_01, subject_s38H/level_L2/rep_09

Of the 2900 successfully detected acquisitions, 933 (32.2%) fall outside the manual-review range of [2, 10] collisions; this splits into 720 (24.8%) with fewer than 2 — typically a clean, easy, or skilled trial, not a defect — and 213 (7.3%) with more than 10, which are the ones actually worth a second look. A breakdown of the 213 high-count acquisitions by phase/level (to check whether they concentrate in Level 5 training, where genuinely more collisions are expected, versus being spread more evenly, which would suggest residual over-detection) is queued (Section 6).

2.4 Instrumentation and feature extraction

Five synchronized data streams were recorded on every trial: electrodermal activity (EDA), eye-tracking (pupil diameter, gaze position), task telemetry (completion time, collision events), tool vibration (two 3-axis accelerometers), and interaction force (a 6-axis force/torque sensor mounted under the task platform). Each stream was processed into its own per-trial feature table ("matrix"), one row per subject × phase × acquisition:

Table (matrix)	Modality	Key columns
eda_features	Electrodermal activity	scl_mean, scr_freq, scr_mean_amp, collision_scr_amp_mean (event-locked)
eye_features	Eye-tracking	pupil_mean, pupil_slope, gaze_dispersion, saccade_rate, collision_pupil_response_mean
perf_features	Task performance	duration_s, n_collisions
force_vibration_features	Force / vibration	tssf_raw (total loading proxy), tssv (total vibration)
merged_features	All four, joined	one row per subject × phase × acquisition present in all four tables

EDA was decomposed into tonic (SCL) and phasic (SCR) components with cvxEDA, a convex-optimization method that avoids the ad hoc peak-detection heuristics of older SCR-scoring approaches (Greco, Valenza, Lanata, Scilingo, & Citi, 2016). Throughout, missing or unreliable data (e.g. a recording gap, no factory calibration for the force sensor) was *excluded and documented*, never interpolated or imputed to fill a gap — data-quality checks describe only what was actually analyzed.

A construct-validity note on tssf_raw: the force sensor is mounted under the task platform, not at the tool tip, so it captures whatever mechanical load transmits through the platform structure during a trial (real task interaction, but also

incidental loading, cross-axis mixing, and platform-transmission effects that cannot be separated with this sensor). It is reported strictly as a *relative, within-study loading proxy*, not a calibrated force in Newtons.

3. Statistical Approach

3.1 A two-layer analysis strategy

Every comparison in this study is now run at two deliberately different levels, and both are reported:

- **Layer 1 — simple, transparent comparisons (Section 4.1).** Each subject contributes one number per phase per outcome (averaged across whatever repetitions/levels fall inside that phase). Within-group paired tests (t-test + Wilcoxon signed-rank) ask "did this group change from Phase A to Phase B"; between-group independent tests (t-test + Wilcoxon rank-sum) ask "do N and H differ, at a phase or in the size of their change"; for the training phase, the same logic extends across the 5 difficulty levels (a Friedman omnibus test per group, plus Level k vs. Level 1 contrasts). No covariates, no random-effects structure, no repeated-measures modeling — a baseline any reader can recompute by hand from the summary tables.
- **Layer 2 — multilevel (mixed-effects) models (Section 4.2 onward).** The full model with Subject as a random-effects grouping factor, fit via fitlme (continuous outcomes, REML) or fitglme (collision counts, Poisson family, log link), using every level-resolved observation rather than a per-phase subject mean. This single-model approach reports all three levels of comparison the design calls for (within-subject slope, between-subjects-within-group variation, between-groups pattern difference) from one fit, with substantially more statistical power than Layer 1 but correspondingly less transparency.

Where the two layers agree, that agreement is one more piece of evidence for a finding. Where they disagree — which happens for the elastic-task GSR transfer effect, Section 4.1.4 — that disagreement is reported explicitly as a caveat on the finding, not silently resolved by reporting only the layer that looks better.

3.2 Key methodological choices and their justification

- **Maximal random-effects structure, with a documented fallback.** The starting structure is a subject-specific intercept and repetition slope allowed to correlate, “keep it maximal” (Barr, Levy, Scheepers, & Tily, 2013). When a fit is degenerate (boundary correlation, near-singular covariance), it is refit with an uncorrelated intercept+slope structure and the non-degenerate fit is reported, following Matuschek, Kliegl, Vasishth, Baayen, and Bates (2017) — which structure was used is recorded and reported alongside every result, not silently decided.
- **Within-subject vs. between-subject disaggregation.** A naive pooled correlation between a physiological measure and performance conflates a trait-level question (“are subjects who are on average more aroused also slower?”) with a state-level question (“on trials where a given subject happens to be more aroused than usual, are they slower on those specific trials?”) — these can have different signs. Person-mean centering separates them explicitly (Curran & Bauer, 2011), which is standard practice for exactly this kind of repeated-measures question.
- **Benjamini-Hochberg FDR correction** (Benjamini & Hochberg, 1995) is applied within each family of related tests (e.g. all primary training-phase interaction tests together), not globally, since unrelated families are not competing for the same error budget.
- **Individual-subject estimation plots, not just group-mean bar/line charts,** for every pre/post comparison, following Weissgerber, Milic, Winham, and Garovic (2015) and Ho, Tumkaya, Aryal, Choe, and Claridge-Chang (2019): a group-mean-only plot can look identical for a uniform small shift across all subjects and for a large change in a few subjects with none in the rest — only individual trajectories distinguish them.
- **Mediation via Monte Carlo confidence intervals** on the product of coefficients (MacKinnon, Lockwood, & Williams, 2004; Selig & Preacher, 2008), the standard approach when an indirect effect's sampling distribution is not itself normal.

- **Every result is labeled confirmatory, pre-specified exploratory, or post hoc exploratory** (Section 5), so a reader is never left inferring how much a given number should be trusted.

4. Results

4.1 Layer 1 — direct pre/post and between-group comparisons

This section reports the simple comparisons described in Section 3.1, across all three comparison scopes the design calls for: the elastic-task transfer probe (Baseline1 vs. Baseline2), the Peg Transfer carry-over control (Test1 vs. Test2), and the training-phase difficulty levels. Full per-outcome tables (11 outcomes × up to 5 comparison types each) are saved alongside this document; the tables below summarize the FDR-corrected significance pattern only — beta/mean-difference magnitudes are in the source .csv files.

4.1.1 Elastic task: Baseline1 → Baseline2

Groups were well-matched at Baseline1 before any haptic exposure — every outcome's between-group comparison at Baseline1 has p_FDR between .78 and .98, i.e. no detectable difference on anything, a formal randomization-balance check.

Outcome	Within N (p_FDR)	Within H (p_FDR)	Between groups, on the change (p_FDR)
scl_mean	**.019 ***	**.0006 ***	.867
scr_freq	**.0002 ***	.072	.867
scr_mean_amp	.166	.192	.905
pupil_mean	**<.0001 ***	**.0002 ***	.905
pupil_slope_norm	.188	.083	.867
gaze_dispersion	.404	.192	.867
saccade_rate	.959	.259	.867
tssf_raw	**.011 ***	**<.0001 ***	.867
tssv	.055	**.014 ***	.867
duration_s	**<.0001 ***	**<.0001 ***	.905
n_collisions	**.011 ***	**<.0001 ***	.867

** = p_FDR < .05. Both groups improve substantially and independently from Baseline1 to Baseline2 on most outcomes (duration_s, n_collisions, pupil_mean, scl_mean, tssf_raw) — a broad, real practice effect that is not haptic-specific. No between-group comparison (at either phase, or on the size of the change) reaches significance for any outcome — see Section 4.1.4 for how this reconciles with the multilevel model's significant scl_mean/scr_freq interaction (Section 4.6).*

4.1.2 Peg Transfer: Test1 → Test2

GSR sample sizes are reduced here (n = 17 N, 16 H, vs. n = 24/22–23 for other modalities) due to the lower GSR coverage at this phase documented in Section 2.3 — interpret the GSR rows with that caveat.

Outcome	Within N (p_FDR)	Within H (p_FDR)	Between groups, on the change (p_FDR)
scl_mean	.341	.088	.960
scr_freq	.341	.354	.960
scr_mean_amp	.341	.653	.976
pupil_mean	**<.0001 ***	**<.0001 ***	.976

Outcome	Within N (p_FDR)	Within H (p_FDR)	Between groups, on the change (p_FDR)
pupil_slope_norm	.697	.653	.976
gaze_dispersion	.341	.354	.496
saccade_rate	.589	.229	.976
tssf_raw	.749	.088	.496
tssv	.697	.287	.960
duration_s	**.0023 ***	**.0025 ***	.976
n_collisions	.367	.680	.960

pupil_mean and duration_s show the same robust within-group improvement seen at Baseline, in both groups. n_collisions stays null within both groups (p_FDR = .37 / .68), consistent with the H3 carry-over-control prediction of no unintended effect on an untrained task. Between-group comparisons are null throughout, the third comparison scope in a row with this pattern (see 4.1.4).

4.1.3 Training-phase difficulty levels

Manipulation check: the 5-level design manipulated difficulty as intended — tssf_raw, tssv, duration_s, and n_collisions all increase sharply and significantly from Level 1 to every other level (2–5), in both groups and pooled (p_FDR typically well below .0001, often below 10^{-15}). This is a clean confirmatory check that the levels differ in actual task demand, independent of any haptic effect.

The N/H asymmetry: scr_freq shows significant level-dependence *within N only* (omnibus p_FDR = .008; a decreasing trend, significant at Levels 3–5) but is flat in H (omnibus p_FDR = .99). **scl_mean** shows the mirror-image pattern: significant level-dependence *within H only* (omnibus p_FDR = .0007; an increasing trend, significant at Levels 2 and 5) but not within N (omnibus p_FDR = .146). Neither outcome's between-group-per-level comparison reaches significance at any individual level (all p_FDR > .81) — the asymmetry shows up as "which group's within-group test clears significance," not as a direct group difference, and is offered as a softer, more transparent version of the Section 4.3 dissociation claim.

pupil_slope_norm (the duration-corrected measure from Section 4.7) shows a genuine, both-groups effect at Level 5 (pooled omnibus p_FDR = .004; Level 5 vs. Level 1 significant in both N, p_FDR = .010, and H, p_FDR = .033) — good corroborating evidence that the corrected measure captures a real signal at the hardest difficulty, not just noise.

As in the other two scopes, between_group_per_level finds no significant group difference at any of the 5 levels, for any outcome (all p_FDR > .81).

4.1.4 Reconciling Layer 1 and the multilevel models

Across all three comparison scopes above, n**ot one between-group test reaches FDR significance** — including for scl_mean and scr_freq at Baseline1→Baseline2, where the Layer 2 multilevel model (Section 4.5) does find a significant Haptic×Phase interaction (p_FDR = .0001 and .0068 respectively). This is reported as an open discrepancy, not resolved in either direction:

- The multilevel model uses every level-resolved observation (n ≈ 425–463 per family) plus Level as a covariate and a proper random-effects structure — substantially more statistical power than Layer 1's subject-level-mean comparison (n ≈ 46–47), and able to detect an effect that is concentrated at specific levels rather than uniform across the phase.
- Averaging across levels before testing, as Layer 1 does, can wash out an effect that depends on level — consistent with the level-dependence already documented for the 3-way training model (Section 4.4).

- Conversely, the fact that the simplest, most direct version of this comparison finds nothing is a legitimate reason for caution: this finding should be reported with the caveat that it depends on the fuller model's extra power and structure, not treated as an unambiguous result that would replicate under any reasonable test.

4.2 Training-phase learning: no differential rate by group (confirmatory, H1/H4)

No outcome — physiological, oculomotor, or behavioral — showed a significant Haptic × Repetition interaction after FDR correction. Both groups improved over the 10 repetitions at the same rate.

Outcome	β (Haptic×Rep)	p
scl_mean	0.0024	.932
scr_freq	-0.0252	.543
pupil_mean	-0.0014	.573
duration_s	-0.037	.842
n_collisions (Poisson)	0.0053	.406

4.3 The dissociation: physiological/oculomotor divergence without a performance analogue (confirmatory, H1/H4)

This is the study's central finding so far. As task difficulty (Level) increases, the gap between groups in physiological and oculomotor arousal grows — but performance measures show no corresponding divergence. For example, SCL's rise from Level 1 to Level 5 is markedly larger in the haptic group (β ≈ 1.32) than the no-haptic group (β ≈ 0.67), while completion time and collisions do not show this group-dependent divergence. In other words, the two conditions differ measurably in the operator's internal state at high difficulty without differing in what the operator actually accomplishes.

Working interpretation (Section 5): this looks less like "haptic feedback reduces physiological cost" (a simple monotonic story) and more like "haptic feedback changes how physiological/attentional resources are allocated as difficulty increases" — the direction and monotonicity of the effect still need to be characterized carefully before choosing between these framings; see 4.4.

4.4 Does the learning-rate difference itself depend on difficulty? (exploratory extension of H1/H4)

The additive model above assumes the group difference in learning rate is the same at every difficulty level. A full three-way test (Haptic × Level × Repetition) shows this assumption does not hold for two outcomes: scr_mean_amp and collision_scr_amp_mean (both p_FDR = .023). Both are driven almost entirely by Level 5: the haptic group shows a sharp negative (habituating) repetition slope specific to the hardest difficulty level, while the no-haptic group shows no such pattern at any level — a plausible **habituation effect**, not a clean "haptic flattens the curve" story across all levels. n_collisions is a near-miss (p_FDR = .053) after re-running with the corrected collision-detection pipeline (Section 6) and is not treated as a finding. This is reported as exploratory.

4.5 Transfer results, multilevel model (confirmatory, H2/H3)

- **Elastic-task transfer (Baseline1→Baseline2):** scl_mean and scr_freq show a significant Phase × Haptic interaction (p_FDR = .0001 and .0068) — the **haptic group**'s skin-conductance measures change more from pre- to post-training. Neither eye-tracking nor performance measures show a corresponding effect. One caveat: without an independent no-training control for the passage of time, randomization supports the group comparison but does not rule out a training-unrelated time effect common to both groups' physiology.

- **Peg Transfer carry-over control (Test1 → Test2):** nothing survives FDR correction — the predicted null result, consistent with the training effects above being specific to tasks resembling the trained one. The collision-detection data-quality reports well-behaved standard errors (SE = 0.09–0.15) and the null carry-over conclusion is unchanged.

4.6 Cross-modal correlation: within- vs. between-subject dissociation (exploratory, pre-specified scope extension)

Provisional: this section's collision-dependent predictors (collision_scr_amp_mean, n_collisions, collision_pupil_response_mean) still reflect the pre-correction collision counts (Section 6) and are queued for re-run; the non-collision predictors are unaffected.

Testing whether physiological/oculomotor state and performance are related trial-to-trial (not just separately affected by group), the within/between disaggregation shows a striking pattern: every within-subject association that reaches significance has a between-subject counterpart that does not, and vice versa. A physiological measure's trial-to-trial coupling with performance and its trait-level relationship are, in this dataset, never the same story — reinforcing that this disaggregation (Curran & Bauer, 2011) is doing real work here, not just a statistical nicety.

4.7 A retracted finding, reported for methodological transparency

An initial analysis found that the within-subject coupling between *pupil_slope* (within-trial pupil-diameter trend) and completion time was extremely strongly moderated by haptic group ($p \approx 3.5\times10^{-19}$). A p-value this extreme warranted checking the estimator itself before treating it as a finding.

The check found the effect was an artifact: *pupil_slope* is computed as a linear-regression slope over each trial's own time window, so its units are mm of pupil change per second, and the regressor's range *is* duration_s — the exact outcome it was being correlated against. A duration-invariant version of the same measure (slope fit against time rescaled to 0–1 within each trial, i.e. mm per fraction of trial) completely eliminated the moderation ($p = .70$ and $.74$ for the two outcomes tested), while the more modest main within-subject association with performance survived.

4.8 Mediation and covariates (confirmatory)

Provisional: the performance outcomes in the mediation test (duration_s, n_collisions) still reflect the pre-correction collision counts for n_collisions (Section 6) and are queued for re-run.

- **Mediation (H5):** no indirect effect of Haptic on either performance outcome through scl_mean, scr_freq, or pupil_mean reached significance (all Monte Carlo 95% CIs span zero; FDR-approximate $p \geq .94$). H5 is not supported by any of the three a-priori candidate mediators.
- **Demographic covariates (SAP Section 6 compliance check):** age, gender, handedness, gaming experience, and prior robotic experience were added to the primary models; the largest coefficient shift was 3.6% (well under the 20% material-change threshold) and no significance status changed. The unadjusted models stand as the primary result.

5. Working Interpretation

The headline pattern — physiological/oculomotor arousal diverges by difficulty and group where performance does not — is easy to misread as “haptic feedback didn't work.” A more accurate reading, consistent with compensatory-control and cognitive-load theory (Hockey, 1997; Kahneman, 1973), is that **operators without haptic feedback may be spending more physiological/attentional effort to hold performance constant**, not failing to hold it constant. However, the non-monotonic level-dependence found in Section 4.4 means this should not yet be framed as a simple “haptic reduces cost” story — the more defensible current framing is that *haptic feedback changes the relationship*

between task difficulty and physiological/oculomotor state, without a specific claim yet about the direction of that change being uniformly cost-reducing.

A terminology note: scl_mean, scr_freq, scr_mean_amp (sympathetic/electrodermal measures), pupil diameter (arousal + luminance + cognitive load), pupil_slope, saccade_rate, and gaze_dispersion (oculomotor scanning behavior) are physiologically distinct signals, not one construct. The write-up avoids grouping them all under a single label like "arousal" and instead interprets each according to its own physiology.

Sample-size framing: although the training phase contributes roughly 2,000+ trial-level observations, the between-group comparison is still driven by N = 47 independent participants (23 vs. 24 per group) — the multilevel model correctly accounts for this in its inference, but it is worth stating explicitly so the trial count is never mistaken for the effective sample size of the group comparison.

On the Layer 1 / Layer 2 discrepancy (Section 4.1.4): every between-group test in the simple, transparent Layer 1 comparisons (Baseline, Peg Transfer, and difficulty levels alike) is null, while the multilevel model finds a handful of significant Haptic-related effects (the scl_mean/scr_freq elastic-task transfer interaction; the level-dependent 3-way effects on scr_mean_amp and collision_scr_amp_mean). Both facts are reported rather than one being allowed to quietly stand in for the whole picture: the multilevel findings are real given the model's assumptions and have more power to detect them, but a reader should know they do not additionally show up in the simplest possible version of the same comparison — that is a genuine limitation on how strongly these particular results should be weighted in the paper's headline claims, not a technicality to bury in a footnote.

6. Current Status and What's Still Being Checked

Several robustness/sensitivity analyses are in progress, following the same principle applied to the pupil-slope finding above — check anything statistically surprising or structurally suspect before reporting it as a headline result.

Item	Status
Collision-count over dispersion	Checked — Pearson dispersion ratio 1.91 (mild over dispersion), but the Haptic×Repetition conclusion is unchanged after correcting for it (p = .41 → .47). Primary result stands.
Completion-time residuals	Checked — raw-scale residuals are non-normal (skew 2.97, kurtosis 24.6), as expected for duration data. A log-scale refit gives much better-behaved residuals and the same (null) conclusion. Recommend reporting the log-scale model as primary.
tssf_raw construct validity	Reframed as "relative platform-interaction loading," not participant-applied force, given the sensor's mounting location.
tssv / haptic-actuator sensor overlap	Investigated and ruled out as a concern for this hardware setup.
saccade_rate as a rate-with-exposure	Poisson-with-offset alternative model built; result pending.
gaze_dispersion vs. duration_s / blink_frac	Correlation checks built; result pending.
Collision-detection audio pipeline	Checked and corrected — the original detector pooled all 6 audiomixer channels through an RMS-smoothed envelope, which over-detected collisions relative to the researcher's own validated visual diagnostic. Replaced with a per-side signed bandpass-sum detector matched to that diagnostic exactly. Affects every collision-dependent outcome (n_collisions, collision_scr_amp_mean, collision_pupil_response_mean); all training-phase (Table 1/2b) and transfer models have been re-run on the corrected counts and are reflected in Sections 4.1, 4.3, and 4.4 above.
Peg Transfer collision-count quality	Checked — Test1/Test2 previously had no working collision detection (source of the earlier degenerate transfer-model fit); now fixed and re-run (Section 4.5).
3-way model interpretation (Section 4.4)	Re-run on corrected collision counts — n_collisions no longer reaches FDR

Item	Status
	significance (p_FDR = .053, was reported significant under the old, over-detecting pipeline). scr_mean_amp and collision_scr_amp_mean remain significant.
Cross-modal correlation (4.6) and mediation (4.8)	Both use collision-dependent columns and still reflect the pre-correction collision counts — re-run pending, results in these two sections should be treated as provisional until then.
Layer 1 simple comparisons (Section 4.1)	Complete for all three scopes (Baseline, Peg Transfer, difficulty levels). See Section 4.1.4 for the open discrepancy with the multilevel models.
Data quality audit (Section 2.3)	Complete — live census against current data, not a description of policy. Two notable findings: s03H/s04H are missing force-sensor data almost entirely (hardware dropout, not task-related), and GSR coverage is lower at Peg Transfer (~82%) than Baseline (~90%) due to that trial's longer, unsegmented recording.
High-count collision review (933 flagged, Section 2.3)	Split into 720 low-count (benign) and 213 high-count (n_collisions > 10, worth review). A breakdown of the 213 by phase/level, to check whether they concentrate in Level 5 training (expected) or are spread more evenly (would suggest residual over-detection), is pending.

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